

Introduction to Electronic Design Automation

Homework 3

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1 BDD Operation

1. (a)

$$\begin{aligned}
 & \text{robdd_build}(ab(\neg c + d) + (a\neg b + \neg ab)(c\neg d + \neg cd), 1) \\
 & \xrightarrow{\eta} \text{robdd_build}(b(\neg c + d) + \neg b(c\neg d + \neg cd), 2) \\
 & \xrightarrow{\eta} \text{robdd_build}(\neg c + d, 3) \\
 & \xrightarrow{\eta} \text{robdd_build}(d, 4) \\
 & \xrightarrow{\eta} \text{robdd_build}(1, 5) \\
 & \quad v_1 \\
 & \xrightarrow{\lambda} \text{robdd_build}(0, 5) \\
 & \quad v_0 \\
 & \quad v_2 = (d, v_1, v_0) \\
 & \xrightarrow{\lambda} \text{robdd_build}(1, 4) \\
 & \quad v_1 \\
 & \quad v_3 = (c, v_2, v_1) \\
 & \xrightarrow{\lambda} \text{robdd_build}(c\neg d + \neg cd, 3) \\
 & \xrightarrow{\eta} \text{robdd_build}(\neg d, 4) \\
 & \xrightarrow{\eta} \text{robdd_build}(0, 5) \\
 & \quad v_0 \\
 & \xrightarrow{\lambda} \text{robdd_build}(1, 5) \\
 & \quad v_1 \\
 & \quad v_5 = (d, v_0, v_1) \\
 & \xrightarrow{\lambda} \text{robdd_build}(d, 4) \\
 & \quad v_2 = (d, v_1, v_0) \\
 & \quad v_4 = (c, v_5, v_2) \\
 & \quad v_6 = (b, v_3, v_4) \\
 & \xrightarrow{\lambda} \text{robdd_build}(b(c\neg d + \neg cd), 2) \\
 & \xrightarrow{\eta} \text{robdd_build}(c\neg d + \neg cd, 3) \\
 & \quad v_4 = (c, v_5, v_2) \\
 & \xrightarrow{\lambda} \text{robdd_build}(0, 3) \\
 & \quad v_0 \\
 & \quad v_7 = (b, v_4, v_0) \\
 & \quad v_8 = (a, v_6, v_7)
 \end{aligned}$$

1. (b)

Since

$$f = ab(\neg c + d) + (a\neg b + \neg ab)(c\neg d + \neg cd)$$

we have

$$f_c = abd + (a\neg b + \neg ab)(\neg d)$$

$$f_{\neg c} = ab + (a\neg b + \neg ab)(d)$$

Using procedure `robdd_build` to f_c , we have

$$\begin{aligned} & \text{robdd_build}(abd + (a\neg b + \neg ab)(\neg d), 1) \\ & \xrightarrow{\eta} \text{robdd_build}(bd + \neg b\neg d, 2) \\ & \quad \xrightarrow{\eta} \text{robdd_build}(d, 4) \\ & \quad \quad v_2 = (d, v_1, v_0) \\ & \quad \xrightarrow{\lambda} \text{robdd_build}(\neg d, 4) \\ & \quad \quad v_5 = (d, v_0, v_1) \\ & \quad \quad v_9 = (b, v_2, v_5) \\ & \xrightarrow{\eta} \text{robdd_build}(b\neg d, 2) \\ & \quad \xrightarrow{\eta} \text{robdd_build}(\neg d, 4) \\ & \quad \quad v_5 = (d, v_0, v_1) \\ & \quad \xrightarrow{\lambda} \text{robdd_build}(d, 4) \\ & \quad \quad v_2 = (d, v_1, v_0) \\ & \quad \quad v_{10} = (b, v_5, v_2) \\ & \quad v_{11} = (a, v_9, v_{10}) \end{aligned}$$

Using procedure `robdd_build` to $f_{\neg c}$, we have

$$\begin{aligned} & \text{robdd_build}(ab + (a\neg b + \neg ab)d, 1) \\ & \xrightarrow{\eta} \text{robdd_build}(b + \neg bd, 2) \\ & \quad \xrightarrow{\eta} \text{robdd_build}(1, 4) \\ & \quad \quad v_1 \\ & \quad \xrightarrow{\lambda} \text{robdd_build}(d, 4) \\ & \quad \quad v_2 = (d, v_1, v_0) \\ & \quad \quad v_{12} = (b, v_1, v_2) \\ & \xrightarrow{\lambda} \text{robdd_build}(bd, 2) \\ & \quad \xrightarrow{\eta} \text{robdd_build}(d, 4) \\ & \quad \quad v_2 = (d, v_1, v_0) \\ & \quad \xrightarrow{\lambda} \text{robdd_build}(0, 4) \\ & \quad \quad v_0 \\ & \quad \quad v_{13} = (b, v_2, v_0) \\ & \quad v_{14} = (a, v_{12}, v_{13}) \end{aligned}$$

Hence, the shared ROBDDs of f , f_c and $f_{\neg c}$ are as shown as below (Figure 1).

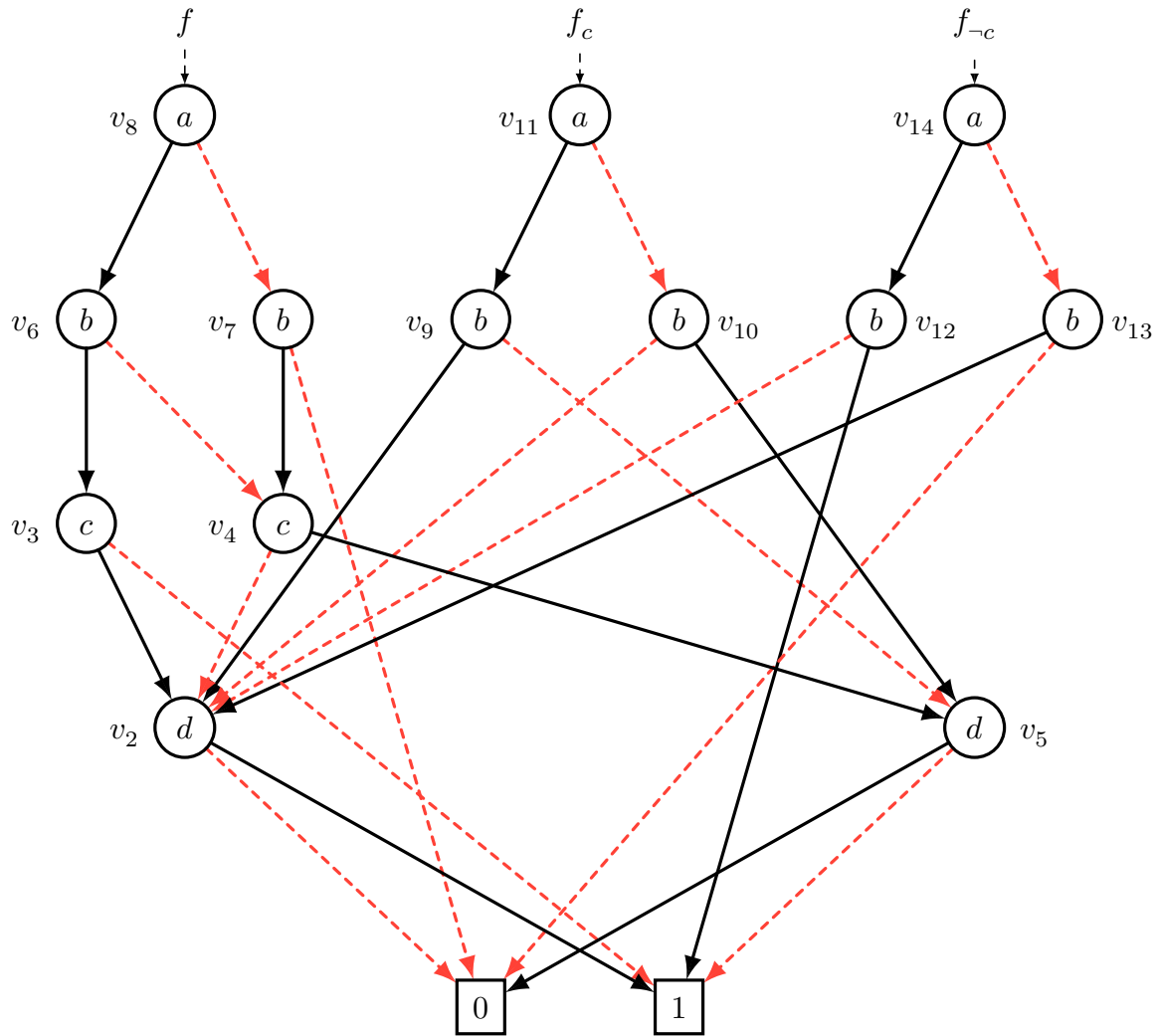


Figure 1: The shared ROBDDs of f , f_c and f_{-c} .
 Red dashed edge represents 0-edge, and black solid edge represents 1-edge.

2 SAT Solving

3 Combinational Equivalence Checking

4 Characteristic Function

5 Sequential Equivalence Checking